

EXPERIENCE

Software Developer Engineer II, Project 990 – Bloomington, IN Jul 2025 – Present

- Achieved 8x throughput on a 5M node & 50M edge graph by adding concurrent traversal starts, caching, and index/memory tuning, demonstrating parallelization and performance optimization for real-time pipelines.
- Implemented CI/CD with GitHub Actions to test and deploy .NET 5 microservices, cutting deploy time 80% and enabling automated hardware-in-the-loop testing workflows.
- Partnered with systems, firmware, and RF engineers to define HW/SW interfaces, document specifications, and validate software on target Linux environments.
- Tech: C, C++, Python, .NET 5 (ASP.NET Core), Docker, Kubernetes, GitHub Actions, Gremlin, AWS, REST, JSON, Linux

Software Engineer, App Orchid – Hyderabad, India Apr 2022 – Aug 2023

- Led 3 engineers to re-architect a low-latency data pipeline for real-time ML inference on embedded SoCs, generating \$1M ARR.
- Ported and optimized signal-processing kernels from MATLAB/Python to C, using SIMD and multi-core parallelism to meet real-time constraints.
- Integrated DSP algorithms with RF, firmware, and systems teams to deploy on custom SoC and validate via hardware-in-the-loop.
- Built CI toolchain (GCC, CMake, Docker) and automated HITL regression tests to support field validation and releases.
- Tech: C, MATLAB, SIMD, SoC integration, GCC, CMake, Docker, Linux, Git, CI.

Software Engineer, HDFC Bank – Bengaluru, India Jul 2020 – Apr 2022

- Ported and optimized inference and signal-processing kernels to C/C++ with SIMD/CUDA optimizations, enabling real-time processing for 50K+ audio calls.
- Engineered batch ETL in Apache Airflow with fan-out/fan-in parallelism and robust retries to maximize deterministic throughput.
- Redesigned database schemas and migrated LINQ to T-SQL stored procedures, reducing query latency by 25% for low-latency pipelines.
- Tech: C, C++, Python, Linux, Git, CMake, Apache Airflow, Docker, Kubernetes, PostgreSQL, Redis, Kafka, Elasticsearch.

Algorithm Software Engineer, Multimodal Robotics Project – San Diego, CA May 2024 – Aug 2024

- Integrated RAG and vision-language models on Nao robot to enable real-time multimodal inference on embedded SoC.
- Built scraping-to-retrieval pipeline using Selenium and FAISS to serve low-latency context for on-device agents.
- Ported Python ML components to optimized C/Python bindings and parallelized pipelines to meet embedded timing constraints.
- Developed hardware-in-the-loop tests and deployment scripts to validate algorithms and collect field data.
- Collaborated with embedded, NLP, and systems engineers; documented interfaces and CI workflows (Django/Flask, chromaDB).

Graduate Research Assistant, Prof. Prateek Sharma – Indiana University Aug 2024 – May 2025

- Ported signal-processing prototypes from MATLAB/C++ to optimized C and CUDA for multi-core SoCs, improving latency 20x and meeting real-time constraints.
- Built low-latency data and control pipelines with gRPC and Dockerized HITL testbeds to validate algorithms on hardware targets.
- Optimized DSP kernels using SIMD/GPU parallelism, multi-threading, and memory-layout tuning to achieve deterministic throughput.
- **Publication:** MQFQ-Sticky — Fair Queueing for Serverless GPU Functions; documented design, tests, and performance tuning.

PROJECTS

- Containerized Real-time Platform | PersonalEngineering
- Containerized React+FastAPI services and deployed to Azure Container Apps / Static Web Apps for scalable real-time workloads.
 - Automated infra with Terraform and CI/CD, standardizing Linux builds and Blob Storage artifacting.
 - Integrated in-browser VS Code to execute code in isolated containers for secure rapid prototyping and hardware-in-the-loop emulation.
 - Built APIs and CI to ingest MATLAB/Python prototypes, versioned via git, and validated via automated tests.

- MERN & FastAPI Backend & CI/CD | Full-StackSoftware Engineering I
- Developed FastAPI microservices for catalogue, booking, and role-based access; deployed on Linux and integrated with git.
 - Implemented CI/CD with GitHub Actions and automated tests to ensure repeatable builds and rapid delivery.
 - Collaborated in Agile/Scrum using Jira with cross-functional engineers to meet tight sprint and field-demo schedules.
 - Authored API specs and deployment documentation to accelerate prototyping and integration with hardware/firmware teams.

SKILLS

Languages	C/C++, Java, Python, JavaScript, TypeScript, SQL
Technology	.NET Core, React, Node, Spring Boot, REST API, Flask, FastAPI, Django, Git, Jira
Data & Tools	AWS, EC2, DynamoDB, S3, SQS, Lambda, Elasticsearch, Spark, Docker, Splunk, Kafka, RabbitMQ

EDUCATION

M.S. in Computer Science, Indiana University, Bloomington, IN | GPA: 3.8/4.0Aug 2023 – May 2025

B.Tech in Aerospace, Minor in Computer Science, Indian Institute of Technology, BombayJul 2016 – Jul 2020